

that the exponential 2^m increases far out there than the quadratic $(m+1)^2$

So $\lim_{m \rightarrow \infty} \frac{2^m}{(m+1)^2}$ does not exist,

the term increases without any upper bound. So the General Term Test

implies. $\sum_{n=1}^{\infty} \frac{2^n}{(n+1)^2}$ is divergent.

Q 10 Applying ratio test to

$$\sum_{n=1}^{\infty} \frac{n}{n+1} = \frac{1}{2} + \frac{2}{3} + \frac{3}{4} + \frac{4}{5}$$

+ ...

this is divergent by the General

Term Test as

$$\lim_{n \rightarrow \infty} \frac{n}{n+1} = 1 \neq 0$$

$$= \lim_{n \rightarrow \infty} \frac{1}{1 + \frac{1}{n}}$$

So hence $\sum \frac{n}{n+1}$ is divergent.

Ratio-test.

$$\frac{k+3}{k^3+2k} \leq \frac{k+3k}{k^3+2k} = 4 \cdot \frac{k}{k^3+2k}$$

$$a > b > 0$$

$$\frac{1}{a} < \frac{1}{b} < 4 \frac{k}{k^3} = 4 \frac{1}{k^2}$$

This shows that

$$0 \leq \frac{k+3}{k^3+2k} < 4 \frac{1}{k^2}$$

$$\text{and } \sum_{k=1}^{\infty} 4 \cdot \frac{1}{k^2} = 4 \sum_{k=1}^{\infty} \frac{1}{k^2}$$

is a known convergent series.

So by the standard comparison test

$$\sum_{k=1}^{\infty} \frac{k+3}{k^3+2k} \text{ converges also.}$$

